Functional Requirements Document (FRD)

Database Implementation Project

1. Project identification

Item	Information
Project title	KOHA-Lite Inventory Management for Little Free Libraries
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Course/section	01:198:437:01
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Version	2.1

2. Business problem and project purpose

Participating Little Free Libraries in New Jersey need a shared database for recording the circulation and location of their books. The libraries currently rely on an honor system, so directors cannot reliably determine which recorded books are available, which copies are on loan, where a returned copy is located, or which titles are in demand.

KOHA-Lite models circulation within participating libraries. Registered patrons and authorized directors record every checkout and return. Inventory represents recorded activity rather than independently verified physical inventory. Directors can reconcile differences between recorded and observed inventory for libraries assigned to them, and every reconciliation retains an attributable audit record.

The project will provide a normalized MySQL database that tracks physical book copies, circulation, donations, visits, director assignments, inventory corrections, and operational reports. It will not include a public website, mobile application, geographic map, cameras, automated physical surveillance, or production identity management.

3. Scope

In scope	Out of scope
Registered fictional users with patron, donor, or director roles	Anonymous or unrecorded borrowing, returns, or donations
Participating libraries and their operational status	Geographic map rendering, route planning, or GPS tracking
Book titles and individually identified physical copies	Book-condition grading, valuation, pricing, fines, or sales
Recorded availability and current library for every copy	Guaranteeing that recorded inventory exactly matches physical inventory
Checkout and mandatory-return workflows	Permanent taking of a checked-out copy
Returns to the originating or another participating library	Waiting lists, reservations, or interlibrary shipping
Donation submission and director acceptance	Purchasing and supplier management
Explicit patron library visits	Passive surveillance or automatic visit detection
Director reconciliation of recorded inventory drift	Public-facing web or mobile application development

In scope	Out of scope
Role-based access, audit history, reports, and database tests	Production authentication, notifications, and enterprise backup automation

4. Stakeholders and user roles

One user may hold multiple roles. A director may manage one or more assigned libraries. Director authority is scoped to assigned libraries; it is not automatically global.

Role	Responsibilities	Database needs
Patron	Records personal checkouts, returns, and visits; reviews personal history	Available-book catalog, own active loans, own borrowing history, own visit history
Donor	Submits donations and reviews aggregate demand information	Donation submission/status and aggregate demand reports
Director	Maintains assigned libraries, records circulation for patrons, accepts donations, and reconciles inventory	Assigned-library inventory, controlled procedures, operational reports, and audit history

Permission matrix

Operation	Patron	Donor	Director
View available books	Yes	Yes	Yes
View aggregate demand	Yes	Yes	Yes
View personal borrowing and visit history	Own records	Only when also a patron	Own records; another patron only for an assigned-library operational task
Record checkout or return	Own activity	Only when also a patron	For patrons at assigned libraries
Record a visit	Own visit	Only when also a patron	For patrons at assigned libraries
Submit a donation	Yes	Yes	Yes
Accept or reject a donation	No	No	Assigned receiving libraries only
Register a physical copy	No	No	Assigned libraries only
Reconcile inventory drift	No	No	Assigned libraries only
Change an assigned library's operational status	No	No	Assigned libraries only
Modify or delete audit history	No	No	No

Roles above are business roles stored in the schema. MySQL roles and grants provide a second enforcement layer for representative database users; the project does not require one MySQL account per patron.

5. Functional requirements

ID	Functional requirement	Priority	Related data/entities
FR-01	An authorized director shall register an individual physical book copy at a library assigned to that director.	High	UserAccount, DirectorLibrary, BookTitle, BookCopy, Library
FR-02	The system shall list available copies at a selected active library, including title, author, category, and available-copy count.	High	Library, BookTitle, BookCategory, BookCopy

ID	Functional requirement	Priority	Related data/entities
FR-03	A registered patron, or a director acting for that patron at an assigned library, shall check out an available copy. A successful checkout shall create one active loan and mark the copy on loan in one transaction.	High	UserAccount, UserRole, BookCopy, Loan, Library
FR-04	A registered patron, or a director acting for that patron, shall return a copy with an active loan. A successful return shall close the loan, mark the copy available, and set its current location to the selected participating return library in one transaction.	High	Loan, BookCopy, Library
FR-05	A registered user shall submit a donation to a selected active library, and an assigned director shall accept or reject it. Accepted physical copies shall be individually registered and linked to the donation.	Medium	Donation, DonationItem, BookTitle, BookCopy, Library
FR-06	A patron, or an assigned director acting for the patron, shall record an explicit library visit with its library, timestamp, and recording actor.	Medium	Visit, UserAccount, Library
FR-07	A patron shall retrieve only that patron's borrowing and visit history. A director may retrieve patron activity only when it concerns a library assigned to that director.	High	Loan, Visit, DirectorLibrary
FR-08	The system shall report demand as the number of distinct patrons who checked out each title during a required date range, optionally filtered to one library, ordered from greatest to least demand.	Medium	Loan, BookCopy, BookTitle, Library
FR-09	An assigned director shall reconcile a recorded copy's location or status when observed inventory differs, supplying a nonblank reason and preserving the prior and new values.	High	InventoryAdjustment, BookCopy, DirectorLibrary
FR-10	The system shall reject unauthorized actions, inactive users, inactive libraries, unavailable-copy checkouts, returns without active loans, and invalid state transitions without leaving partial circulation or inventory changes.	High	All transactional entities
FR-11	The system shall create append-only audit evidence for successful state-changing business operations, including actor, event, affected record, timestamp, and relevant context.	High	AuditEvent
FR-12	An assigned director shall mark that library active or inactive through a controlled operation. Inactive libraries shall retain history but shall not accept new checkout, return, visit, donation, or copy-registration activity.	Medium	Library, DirectorLibrary, AuditEvent
FR-13	A repeated checkout or return request that conflicts with the current state shall be rejected without creating another loan, return, or inventory change.	High	Loan, BookCopy, AuditEvent

6. Data requirements

Data subject	Information to store	Identifier
User account	Synthetic display name, synthetic email, active status, creation time	UserID
Role assignment	User and business role	UserID + RoleCode
Library	Name, address, operational status	LibraryID
Director assignment	Director and managed library	UserID + LibraryID
Book category	Category name and description	CategoryID

Data subject	Information to store	Identifier
Book title	ISBN when available, title, author, category	BookTitleID
Book copy	Title, current library, status, registration source and time	BookCopyID
Loan	Copy, patron, checkout library/actor/time, return library/actor/time, state	LoanID
Donation	Donor, receiving library, submission state, decision actor/time	DonationID
Donation item	Donation, title, offered quantity, accepted quantity	DonationItemID
Visit	Patron, library, visit time, recording actor and method	VisitID
Inventory adjustment	Copy, director, prior/new status and location, reason, time	AdjustmentID
Audit event	Actor, event type, entity, record identifier, timestamp, outcome, context	AuditEventID

All identities and contact values used for implementation and demonstration are synthetic. Price is not stored because the system does not perform valuation, purchasing, fines, insurance, or sales. Availability and quantity are derived from individual copy records rather than maintained as an independently editable quantity.

7. Business rules and validation rules

ID	Business rule	Planned database enforcement
BR-01	Every persistent record has a unique identifier; role and assignment pairs cannot be duplicated.	Primary keys and UNIQUE constraints
BR-02	Every physical copy belongs to exactly one book title and has at most one current library.	NOT NULL title FK; nullable library FK only for retired/lost states
BR-03	A copy may have at most one active loan.	Generated active-copy key with UNIQUE constraint plus controlled procedure
BR-04	Only an AVAILABLE copy at the selected active library may be checked out.	Transactional procedure, locking read, status checks
BR-05	A checkout and its copy-status change succeed or fail together.	InnoDB transaction with rollback handler
BR-06	A return must reference an active loan, and a returned copy becomes AVAILABLE at the participating return library.	Transactional procedure and state validation
BR-07	A patron must be active and have the PATRON role before a checkout, return, or visit is recorded for that patron.	Role-check function and controlled procedures
BR-08	A director may register copies, accept donations, record patron activity, or reconcile inventory only for an assigned library.	Director-assignment function and controlled procedures
BR-09	Donation offered quantity must be positive; accepted quantity must be between zero and offered quantity.	CHECK constraints and acceptance procedure
BR-10	Demand is COUNT(DISTINCT PatronUserID) over successful checkouts in a required date range.	Parameterized report query
BR-11	An inactive library retains existing records but cannot receive new operational activity.	Library-status checks in controlled procedures
BR-12	Inventory reconciliation requires an actor, affected copy, prior/new values, and nonblank reason.	NOT NULL/CHECK constraints and reconciliation procedure
BR-13	Audit events are append-only.	Privileges plus triggers that reject UPDATE and DELETE
BR-14	Completed circulation history is not overwritten or hard-deleted.	No direct DML grants; append/close workflow through procedures

ID	Business rule	Planned database enforcement
BR-15	A failed or repeated circulation request leaves the database in its prior consistent state.	State checks, unique active-loan constraint, rollback handler

8. Reports and queries

ID	Report/query name	Purpose	Required result
RQ-01	Available catalog	Show what patrons can borrow at a selected library	Library and optional category filter; title, author, category, available-copy count; title order
RQ-02	Patron active loans	Show a patron's currently borrowed copies	Authorized patron; copy ID, title, checkout library and time; oldest checkout first
RQ-03	Patron borrowing history	Show a patron's completed and active circulation	Authorized patron and optional date range; copy/title, checkout and return timestamps, libraries, loan state
RQ-04	Title demand	Identify titles used by the most distinct patrons	Required start/end timestamps and optional library; title, distinct-patron count, checkout count; descending demand
RQ-05	Patron visit history	Show explicit visits recorded for a patron	Authorized patron and optional date range; library, visit timestamp, recording method; newest first
RQ-06	Assigned-library inventory	Show recorded inventory for a director's assigned library	Director and assigned library; copy, title, recorded location, copy status, active-loan indicator
RQ-07	Donation status	Show submission and decision progress	Donor or assigned director; donation, receiving library, items, offered/accepted counts, status and decision time
RQ-08	Audit history	Review attributable changes for an authorized scope	Director and assigned library or affected record; actor, event type, timestamp, entity/ID, result and context

9. Assumptions and constraints

- All circulation, returns, donations, and explicit visits covered by the system are recorded; anonymous activity is outside the modeled workflow.
- A checked-out copy is expected to be returned. The project does not impose a due date, overdue fine, or notification because none was requested.
- A return may occur at a participating library different from the checkout library.
- Recorded inventory can drift from observed physical inventory; assigned directors correct drift through an audited reconciliation operation.
- One user may have multiple business roles, and one director may be assigned to multiple libraries.
- Library visits are explicit user/director-recorded events, not inferred from loans and not captured through surveillance.
- The database is implemented for MySQL 8.0 or later using InnoDB and UTF-8 (`utf8mb4`).

- The implementation uses fictional sample identities and does not contain real personal information or credentials.
- A public UI, production authentication service, production backup automation, and physical inventory verification are outside scope.

10. Acceptance criteria

Requirement	Observable acceptance criterion
FR-01	An assigned director registers a copy successfully; an unassigned director's attempt is rejected with no copy inserted.
FR-02	With a known fixture, the selected library catalog returns the expected titles and available-copy counts and excludes on-loan copies.
FR-03	Checkout of an available copy creates exactly one active loan and changes the copy to ON_LOAN in one transaction. Checkout of an unavailable copy changes nothing.
FR-04	Return closes the active loan and makes the copy available at the chosen participating library. A duplicate return changes nothing.
FR-05	A donor submits a donation; an assigned director accepts it; accepted counts and linked individual copies match the approved quantities.
FR-06	Patron self-check-in and director-recorded check-in create visits with the correct patron, library, actor, time, and recording method.
FR-07	A patron can retrieve that patron's history and cannot retrieve another patron's private history through the protected interface.
FR-08	For a fixture containing known checkouts, the demand report returns the expected distinct-patron count for the supplied date range and library filter.
FR-09	An assigned director changes recorded location/status with a reason; the copy and immutable adjustment history show the expected prior and new values.
FR-10	Unauthorized, invalid-state, and inactive-library operations return errors and leave related tables unchanged.
FR-11	Each successful state-changing operation creates an audit event containing actor, event type, timestamp, entity/ID, and context. UPDATE or DELETE of an audit event is rejected.
FR-12	After a library becomes inactive, history remains queryable while new operational transactions for that library are rejected.
FR-13	Two requests for one available copy result in at most one active loan; retrying a completed return creates no extra return or inventory increment.

Traceability note: Every FR requirement is mapped in the TDD to tables, constraints, procedures, reports, tests, and reproducible validation evidence. The full executable definition is maintained in `database/`, and the authoritative build order is `database/manifest.txt`.

11. Revision history and feedback disposition

Document revision history

Version	Revision	Summary
1.0	Initial draft	Established the Little Free Library topic, initial scope, three functional requirements, three business rules, three reports, assumptions, and acceptance criteria.

Version	Revision	Summary
2.0	Requirements revision	Reworked the FRD after instructor feedback and project-scope clarification. Added explicit workflows, physical-copy tracking, permissions, reporting definitions, auditability, transaction rules, and testable acceptance criteria.
2.1	Feedback traceability revision	Added this revision history and documented which feedback was adopted, adapted, or not adopted, including the reason for each decision. No business behavior changed from version 2.0.

Feedback disposition

This table follows the section headings and recommendations in the instructor feedback document. `Adopted` means the recommendation was added substantially as proposed. `Adapted` means the concern was addressed with a project-specific implementation. `Not adopted` identifies a proposed permission or feature that the feedback made conditional and that was excluded for a stated reason.

Feedback topic	Disposition	Revision and rationale
Clarify the operating model	Adopted	Sections 3, 5, 7, and 9 now identify who records checkout and return activity, require registered and attributable actors, define return behavior, allow a participating return location, distinguish recorded from observed inventory, and provide audited reconciliation. These revisions remove ambiguity about how an honor-system library produces reliable database records.
Turn roles into entitlements	Adapted	Section 4 adds a permission matrix for patrons, donors, and directors; supports multiple business roles and multiple director-library assignments; and limits director actions to assigned libraries. Business roles are enforced through schema relationships and controlled operations, with representative MySQL privileges documented in the TDD.
Director access to another patron's contact details	Not adopted	The proposed permission was conditional on explicit justification. No retained requirement needs directors to retrieve another patron's contact details, so that access was excluded to keep private data outside operational reports.
Expand and rewrite the functional requirements	Adopted	The three broad original requirements were replaced by thirteen numbered, testable requirements covering every retained capability: copy registration, catalog access, circulation, donations, visits, private history, demand reporting, reconciliation, authorization, audit evidence, library status, and retry safety. This supports direct traceability to design and tests.
Distinguish book titles from physical copies	Adopted	The model separates shared `BookTitle` data from individual `BookCopy` records so copies of one title can have different locations and states. Price was removed because the project performs no valuation or sales; quantity is derived from copy rows; visits are recorded explicitly; and library-to-book collections are modeled as relationships.
Define workflows and business-rule enforcement	Adopted	Checkout, return, donation, and reconciliation now specify actors, preconditions, state changes, rejection behavior, and audit results. Availability is enforced by transactional procedure logic rather than foreign keys alone. A locking read and unique active-copy key address simultaneous checkout requests, and failed operations roll back without partial changes.

Feedback topic	Disposition	Revision and rationale
Make reports precise enough to drive queries and indexes	Adopted	Section 8 defines the purpose, inputs, output columns, filters, aggregation, and ordering for each report. Demand uses a required date window and a defined distinct-patron measure, with checkout count retained as supporting context. These definitions are specific enough to implement and verify in SQL.
Replace the acceptance criteria with observable tests	Adopted	Section 10 replaces map rendering, cryptographic/dataflow language, and catalog-based ACID claims with setup/action/outcome tests for availability, concurrent checkout, return, unauthorized history access, known report fixtures, rollback, retries, reconciliation, and immutable audit evidence. The tests now measure the requirements they claim to validate.
Add auditability and resiliency requirements	Adapted	Successful state-changing operations require attributable, append-only audit evidence; transactions roll back incomplete work; repeated requests are rejected safely; inventory drift is corrected through recorded reconciliation; and the build initializes synthetic data reproducibly. Failed-attempt logging is kept separate from successful business-event history, as the feedback recommends.
Clean up inconsistent assumptions and terminology	Adopted	The unrelated supplier assumption was removed; `Reader` and `Customer` were standardized to `Patron`; demonstration identities and contact values are explicitly synthetic; library status is defined as operational `ACTIVE` or `INACTIVE`; and generic report labels were replaced by concrete results.